

Determinants and maximal-minor factorization

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Scope

Determinant differences and maximal minors turn coordinate changes into controlled algebraic data. Cramer coordinates and the uniqueness of a right factor will later translate matching minor vectors into a linear map.

7 direct Cards + 2 reused prerequisites = 9 unique Cards. This count is a selected Card closure, not a source-declaration count.

[Route reading PDF](#) · [Preserved source exploration](#)

Cards in this route

- Key result: [A unique right factor from proportional oriented maximal minors](#)
- Key result: [Cramer's rule for coordinates of a row](#)

[Read this route with prerequisites](#)

Dependency-first reading route

Read selected Card prerequisites before their uses. Levels are recomputed from the selected reachability-preserving projection; omitted source helpers remain traceable in the source exploration.

0 1

Level 0 (5 Cards)

Level 0

The family of maximal minors in increasing row order

Collects the determinants of all square submatrices with all columns and an increasing selection of rows.

`MathlibAnnex.Matrix.maximalMinors`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [The satellite polynomial is linear in the maximal-minor vector](#) · [Maximal minors scaled by the reference ball volume](#) · [Right multiplication scales every maximal minor by one determinant](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

A determinant difference bound in the sup operator norm

Controls determinant variation by replacing one matrix row at a time.

`MathlibAnnex.ContinuousLinearMap.abs_det_sub_le_max`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)

Level 0

Cramer's rule for coordinates of a row

Expresses a row in the basis of the rows of an invertible square matrix through row-replacement determinants.

`MathlibAnnex.Matrix.det_smul_vecMul_nonsingInv_eq_updateRowDet`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [A unique right factor from proportional oriented maximal minors](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

A weighted determinant polynomial with satellites

Adds linear row-replacement contributions to a base determinant.

`MathlibAnnex.Satellite.satellitePolynomial`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [The satellite polynomial is linear in the maximal-minor vector](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

A seminorm with two-sided bounds against a reference norm

Records a norm through a seminorm, two positive comparison constants and continuity on the original normed space.

`MathlibAnnex.EquivalentSeminorm`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Maximal minors scaled by the reference ball volume](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1 (4 Cards)

Level 1

The satellite polynomial is linear in the maximal-minor vector

Expresses the configuration polynomial as a linear pairing with maximal minors.

`MathlibAnnex.PluckerSupport.pluckerPairing_satelliteCoefficients_maximalMinors`

Immediate Card prerequisites: [The family of maximal minors in increasing row order](#) · [A weighted determinant polynomial with satellites](#)

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)

Level 1

Maximal minors scaled by the reference ball volume

Fixes the volume factor and increasing-row signs.

`MathlibAnnex.Matrix.ballVolumeScaledMaximalMinors`

Immediate Card prerequisites: [The family of maximal minors in increasing row order](#) · [A seminorm with two-sided bounds against a reference norm](#)

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)

Level 1

A unique right factor from proportional oriented maximal minors

Recovers a square factor and its determinant from proportional determinants of every ordered row tuple.

`MathlibAnnex.Matrix.existsUnique_factor_of_orientedMaximalMinorsProportional`

Immediate Card prerequisites: [Cramer's rule for coordinates of a row](#)

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)

Level 1

Right multiplication scales every maximal minor by one determinant

Turns a square right factor into a common scalar on the family of maximal minors.

`MathlibAnnex.Matrix.maximalMinors_mul`

Immediate Card prerequisites: [The family of maximal minors in increasing row order](#)

Used by in this scope: None in this selected scope

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